

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	Joanathan Packia Singh	Reg. No.:	192472229
Date:	22/08/2026	Duration:	60 minutes

Code of Conduct

I Joanathan Packia Singh (192472229) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Joanathan

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25



SIMATS ENGINEERING

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CO5 ASSESSMENT TOOL 1 — RESEARCH PAPER REVIEW / LITERATURE SURVEY

Fitness and Diet Tracker with Personalized Recommendations

Course Code	CSA10 / CSA1063
Course Title	Software Engineering
CO Assessed	CO5 — Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly
Assessment	Assessment Tool 1 — Research Paper Review / Literature Survey
Weightage / Total Marks	20% / 25 Marks
Student Name	Joanathan Packia Singh
Register Number	192472229
Assigned Problem No.	Problem 22 — Fitness and Diet Tracker with Personalized Recommendations
Papers Reviewed	8 papers, all published 2023–2026
Duration	60 minutes

Code of Conduct

I, Joanathan Packia Singh (Reg. No. 192472229), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Joanathan

TABLE OF CONTENTS

1. Introduction and Problem Context	3
2. Literature Search Methodology	3
3. Individual Paper Reviews.....	4
4. Comparative Analysis	7
5. Limitations and Challenges of Existing Approaches	8
6. Research Gap Analysis.....	9
7. Future Research Directions	10
8. Comparative Literature Survey Table.....	10
9. Conclusion	11
References.....	12

1. Introduction and Problem Context

The assigned problem (Problem 22) is a Fitness and Diet Tracker with Personalized Recommendations — a platform that monitors physical activity, calorie intake, and nutritional habits, then converts that data into personalized workout and diet guidance, tracks progress, integrates wearable devices, and sends reminders while protecting user data. This literature survey reviews eight peer-reviewed research papers, all published between 2023 and 2026, that address the core technical components of this problem: exercise recommendation, nutrition recommendation, and wearable-based health monitoring. The goal is to understand existing approaches and technologies, compare them on relevant parameters, identify unresolved research gaps, and propose future directions that could inform the continued development of FitTrack Pro, the system designed across this course's earlier CO3 and CO4 assessments.

2. Literature Search Methodology

2.1 Search Strategy

Papers were identified using Google Scholar, ACM Digital Library, IEEE Xplore, ScienceDirect, and publisher databases (Nature/Scientific Reports, MDPI), using search terms including “personalized exercise recommendation machine learning”, “AI diet recommendation deep learning”, “personalized nutrition recommendation review”, and “wearable IoT health monitoring deep learning”. Search results were filtered to peer-reviewed journal articles and systematic reviews published between January 2023 and 2026, directly relevant to fitness tracking, diet recommendation, or wearable-based activity/health monitoring.

2.2 Inclusion Criteria

- Published between 2023 and 2026 (all eight selected papers satisfy this).
- Directly relevant to at least one FitTrack Pro module: exercise/activity recommendation, nutrition/diet recommendation, or wearable sensing and health monitoring.
- Peer-reviewed journal article, systematic review, or ACM/IEEE-indexed publication with a verifiable DOI.
- Reports a concrete methodology, dataset, or technology stack (not an opinion piece or non-technical article).

2.3 Selected Papers Overview

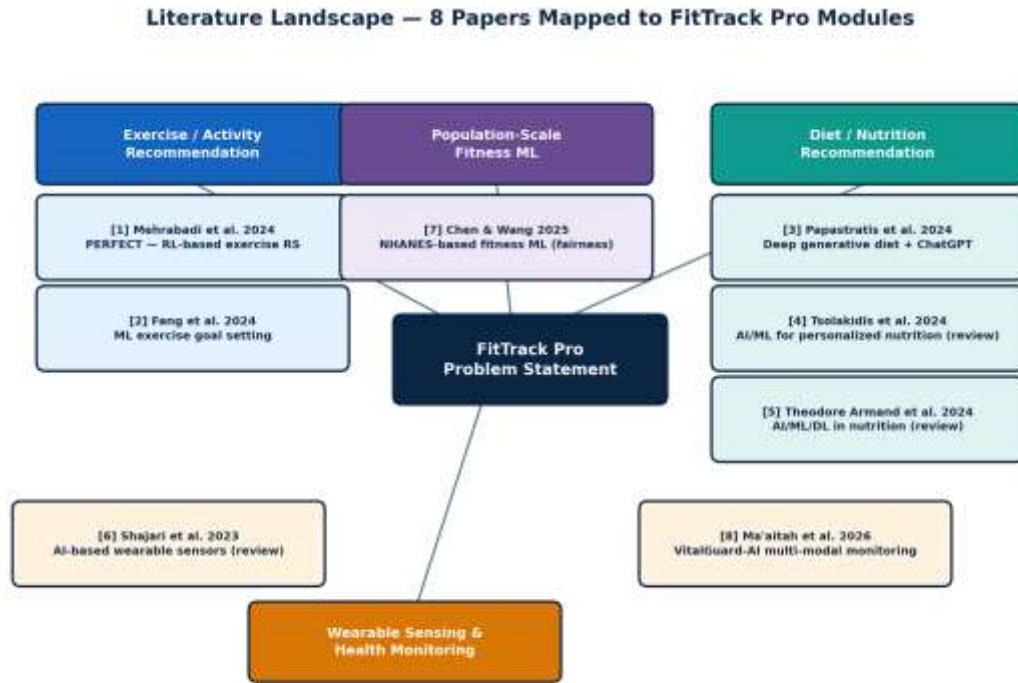


Figure 2.1 — The eight reviewed papers clustered by FitTrack Pro module

3. Individual Paper Reviews

Each paper below is summarized in a consistent format — objectives, methodology, technologies/dataset, and key findings — directly from the published abstract and results, with no invented figures or claims.

3.1 [1] Mehrabadi et al. (2024) — PERFECT: Personalized Exercise Recommendation Framework and archiTecture

Aspect	Summary
Objective	Enhance physical activity levels by developing a personalized exercise recommendation system that proposes tailored exercise plans based on biomarkers and user context.
Methodology	Reinforcement learning via a contextual multi-armed bandit algorithm; smartphone and smartwatch applications gather, monitor, and recommend exercise routines in real time.
Technologies / Dataset	Wearable trackers, IoT, and mHealth integration; real-time heart-rate monitoring; evaluated with 20 female college students over a closed-loop mHealth study.
Key Findings	Significant increase in average daily exercise duration ($P < .001$); high participant satisfaction (4.31/5 for the walking program, 3.69/5 for the recommendation component).

3.2 [2] Fang, Lee, Ji & Wang (2024) — Enhancing Digital Health Services: A Machine Learning Approach to Personalized Exercise Goal Setting

Aspect	Summary
Objective	Propose a machine learning methodology for personalized exercise goal setting in digital health, focused on user exercise behaviour rather than one-size-fits-all daily targets.
Methodology	A knowledge-based digital health system built around an exercise performance evaluation model that selects goals while modeling the effects of fitness and fatigue.
Technologies / Dataset	Exercise performance and fitness profile data; no wearable hardware proposed — focuses on the goal-setting algorithm layer above existing activity data.
Key Findings	Fixed daily exercise plans were found to demotivate users; personalized, fatigue-aware goal setting is proposed as a more sustainable alternative to static targets.

3.3 [3] Papastratis, Konstantinidis, Daras & Dimitropoulos (2024) — AI Nutrition Recommendation Using a Deep Generative Model and ChatGPT

Aspect	Summary
Objective	Introduce an AI-based nutrition recommendation method that aligns generated meal plans with established nutritional guidelines, addressing trustworthiness concerns in AI dietary advice.
Methodology	A variational autoencoder models anthropometric measurements and medical conditions in a latent space; an optimizer adjusts meal quantities to energy requirements; ChatGPT supplies meal variety across cuisines.
Technologies / Dataset	Deep generative network + large language model integration; evaluated on 3,000 virtual user profiles (84,000 daily meal plans) and 1,000 real profiles (7,000 daily meal plans).
Key Findings	The method achieves high accuracy, nutritional appropriateness, and meal variety, demonstrating that combining a generative network with an LLM can outperform either approach alone.

3.4 [4] Tsolakidis, Gymnopoulos & Dimitropoulos (2024) — Artificial Intelligence and Machine Learning Technologies for Personalized Nutrition: A Review

Aspect	Summary
Objective	Survey data-driven AI/ML technologies and data collection technologies for personalized nutrition, and identify open research challenges.
Methodology	PRISMA-guided systematic literature review across three databases (Scopus, ScienceDirect, IEEE Xplore), covering 2021–2024, screening 3,052 records down to 67 final studies.
Technologies / Dataset	Covers nutrition, recipe, and restaurant recommendation systems (43 works) and wearable/camera/smartphone/other data-collection technologies (17 works).
Key Findings	Recommendation systems dominate the personalized-nutrition field; identifies unresolved challenges in data integration, precision/accuracy, genetic variation, long-term effects, behaviour change, and privacy/ethics.

3.5 [5] Theodore Armand, Nfor, Kim & Kim (2024) — Applications of AI, ML, and DL in Nutrition: A Systematic Review

Aspect	Summary
Objective	Comprehensively investigate the current landscape of AI/ML/DL applications across nutrition science broadly, beyond just meal recommendation.
Methodology	Hybrid approach combining systematic literature review (SLR) guidelines and PRISMA guidelines to analyze the scientific literature from major databases.
Technologies / Dataset	Reviews applications spanning smart/personalized nutrition, dietary assessment, food recognition and tracking from images, and predictive modeling for disease prevention/monitoring.
Key Findings	AI/ML/DL show versatility in handling complex nutritional-dataset relationships; the review highlights both significant promise and remaining implementation challenges for real-world deployment.

3.6 [6] Shajari, Kuruvinashetti, Komeili & Sundararaj (2023) — The Emergence of AI-Based Wearable Sensors for Digital Health Technology: A Review

Aspect	Summary
Objective	Review recent progress in wearable sensor hardware (physical, chemical, biosensors) and how AI is combined with wearable technology for intelligent sensing.
Methodology	Structured literature review covering sensor materials, structural configurations, and transduction mechanisms, paired with AI techniques for big-data processing and real-time analysis.
Technologies / Dataset	Physical/chemical/biosensor hardware; AI methods for self-learning, power-efficient, real-time on-device processing enabling personalized health monitoring.
Key Findings	AI-enabled wearables show strong potential for personalized, continuous health monitoring, but accuracy limitations and diagnostic precision remain open engineering challenges.

3.7 [7] Chen & Wang (2025) — Personalized Fitness Recommendations Using Machine Learning for Optimized National Health Strategy

Aspect	Summary
Objective	Develop an ML framework generating personalized fitness recommendations at population scale while maintaining fairness across demographic subgroups, aligned with national health goals.
Methodology	Supervised regression (weekly activity minutes) and classification (risk tiers) using XGBoost, Decision Trees, and Artificial Neural Networks; k-means clustering for health-profile personalization; risk-based goal scaling.
Technologies / Dataset	NHANES 2017–2020 (~15,000 participants) enriched with BRFSS behavioural/environmental variables (mental health days, access to exercise facilities).
Key Findings	XGBoost achieved the best performance (MeanIoU 0.789, F1 > 0.79 across all risk tiers) with fairness gaps below 0.05 across age, gender, and ethnicity — and a lower MAE (28.9 min/week) than prior comparable NHANES-based models.

3.8 [8] Ma’aitah, Idoko & Yaseen (2026) — VitalGuard-AI: A Real-Time Multi-Modal Deep Learning Framework for Intelligent Health Monitoring Using Wearable IoT Devices

Aspect	Summary
Objective	Address real-time processing and multi-sensor data fusion limitations in existing wearable IoT health-monitoring systems.
Methodology	A Multi-Modal Adaptive Health Network (MAHN) combining CNNs and LSTM networks to fuse photoplethysmography (PPG), ECG, accelerometer, and temperature signals, deployed on an edge-cloud hybrid architecture for real-time inference.
Technologies / Dataset	Wearable IoT multi-sensor devices; evaluated on data from 3,247 participants collected over an eight-month period.
Key Findings	High accuracy across five monitoring tasks: cardiac anomaly detection (98.7%), sleep pattern analysis (97.2%), stress classification (95.8%), and activity recognition (94.3%), demonstrating the viability of real-time multi-modal fusion on wearable hardware.

4. Comparative Analysis

The eight papers are compared below on dimensions directly relevant to FitTrack Pro's requirements: reported accuracy/performance, scalability (sample size and deployment scope), and methodology type.

Paper	Methodology	Scale	Reported Performance
[1] Mehrabadi et al. 2024	RL / contextual bandit	n = 20 (small pilot)	P < .001 improvement in exercise duration; 4.31/5 satisfaction
[2] Fang et al. 2024	Knowledge-based + performance evaluation model	Not population-scale (framework paper)	Qualitative improvement over fixed daily plans; no large-scale accuracy metric reported
[3] Papastratis et al. 2024	Variational autoencoder + LLM (ChatGPT)	3,000 virtual + 1,000 real profiles (84,000 + 7,000 meal plans)	High accuracy, variety, and generalization across both virtual and real profiles
[4] Tsolakidis et al. 2024	PRISMA systematic review	67 studies synthesized (from 3,052 screened)	Meta-level: maps the field rather than reporting a single accuracy figure
[5] Theodore Armand et al. 2024	Hybrid SLR + PRISMA systematic review	Field-wide synthesis (Nutrients journal)	Meta-level: breadth across smart nutrition, dietary assessment, food recognition, disease prediction
[6] Shajari et al. 2023	Structured hardware + AI review	Field-wide synthesis (Sensors journal)	Meta-level: strong potential noted, alongside explicit accuracy/diagnostic-precision limitations
[7] Chen & Wang 2025	XGBoost / Decision Tree / ANN, k-means clustering	~15,000 NHANES participants (population-scale)	MeanIoU 0.789, F1 > 0.79, fairness gap < 0.05, MAE 28.9 min/week (best vs. prior NHANES models)

Paper	Methodology	Scale	Reported Performance
[8] Ma'aitah et al. 2026	CNN + LSTM multi-modal fusion (MAHN)	3,247 participants over 8 months	98.7% cardiac anomaly, 97.2% sleep, 95.8% stress, 94.3% activity recognition accuracy

4.1 Observations from the Comparison

- Scale varies enormously: from a 20-person pilot [1] to population-scale datasets of ~15,000 [7] and 3,247 [8] participants — FitTrack Pro should treat small-cohort results (like [1]) as directional rather than production-ready accuracy claims.
- The two review papers on nutrition ([4], [5]) converge on the same conclusion from largely independent literature sets: recommendation systems are the dominant technical approach, which supports FitTrack Pro's own recommendation-engine-centred design.
- The highest quantitative accuracy figures come from narrowly-scoped classification/detection tasks ([7]'s risk tiers, [8]'s cardiac/sleep/stress detection) rather than from open-ended recommendation quality, which is inherently harder to benchmark numerically — a pattern FitTrack Pro's own evaluation strategy should anticipate.
- Only [3] and [7] explicitly evaluate on both a large synthetic/population dataset and report generalization behaviour, making them the strongest methodological references for FitTrack Pro's own recommendation engine and dashboard analytics respectively.

5. Limitations and Challenges of Existing Approaches

Limitation / Challenge	Evidence from Reviewed Literature
Small sample sizes in pilot studies	[1] Mehrabadi et al. 2024 evaluates on only 20 participants, all female college students — limiting generalizability across age, gender, and fitness-level diversity.
Fixed, non-adaptive goal setting	[2] Fang et al. 2024 explicitly identifies that many existing digital-health systems still use fixed daily targets that demotivate users rather than adapting to fatigue and performance.
AI trustworthiness in nutrition advice	[3] Papastratis et al. 2024 notes that a lack of guidelines from nutritional experts has raised real questions about the accuracy and trustworthiness of AI-generated nutrition advice.
Fragmented data collection standards	[4] Tsolakidis et al. 2024 identifies the integration of diverse, multimodal datasets (physiological, genetic, preference) as an unresolved challenge with no standardized protocol.
Limited genetic/precision personalization	[4] found only 3 of 67 reviewed nutrition studies incorporated genetic data, despite genetic variation strongly influencing individual dietary response.
Short study durations	[4] notes most human studies span only a few months, insufficient to establish long-term health outcome effects of personalized recommendations.
Hardware accuracy and diagnostic precision	[6] Shajari et al. 2023 finds that despite progress, wearable sensors still have limitations in data accuracy and precise disease diagnosis, particularly for chemical/biosensors.
Demographic underrepresentation	[7] Chen & Wang 2025 found slightly lower model performance for Black and other underrepresented racial/ethnic groups in the NHANES dataset, despite an overall low fairness gap.
Cross-sectional data limits longitudinal claims	[7] is built on cross-sectional NHANES data, limiting its ability to model behavioural change or long-term outcomes over time — the authors flag this explicitly as a limitation.

Limitation / Challenge	Evidence from Reviewed Literature
Real-time fusion resource cost	[8] Ma'aitah et al. 2026's edge-cloud hybrid model improves real-time responsiveness but implies ongoing computational/connectivity cost for continuous multi-sensor fusion, a constraint for low-cost wearable deployment.

6. Research Gap Analysis

Synthesizing the eight reviewed papers against FitTrack Pro's problem statement — which requires activity tracking, nutrition tracking, personalized recommendations, wearable integration, progress dashboards, reminders, and secure data handling in a single unified platform — reveals several specific gaps that no single reviewed paper closes on its own.

6.1 Gap 1: No Unified Exercise-and-Nutrition Recommendation Model

Papers [1] and [2] address exercise recommendation; papers [3], [4], and [5] address nutrition recommendation — but none of the eight papers proposes or evaluates a single model that jointly optimizes exercise and diet recommendations against the same user profile. FitTrack Pro's problem statement explicitly requires both, meaning this integration itself is a research gap, not just an engineering task.

6.2 Gap 2: Fairness Evaluation Is Present but Not Universal

Only [7] Chen & Wang 2025 conducts an explicit demographic fairness audit (age, gender, ethnicity). None of the exercise-recommendation papers [1, 2] or the wearable-monitoring papers [6, 8] report subgroup performance breakdowns, despite personalization being their central claim — a personalization system that hasn't been checked for demographic bias cannot fully substantiate that claim.

6.3 Gap 3: Short-Term Studies Dominate the Evidence Base

[1]'s pilot ran four weeks; [4] notes most reviewed nutrition studies span only a few months. No reviewed paper reports outcomes beyond roughly one year. Whether personalized recommendations from any of these approaches produce durable behaviour change — the actual goal of a fitness and diet tracker — remains empirically unresolved.

6.4 Gap 4: Real-Time Multi-Modal Fusion Is Proven for Vitals, Not Yet for Recommendation Loops

[8] demonstrates that real-time multi-modal sensor fusion (PPG, ECG, accelerometer, temperature) can achieve high accuracy for detection tasks (cardiac anomaly, sleep, stress). However, no reviewed paper closes the loop by feeding this real-time physiological stream directly into an exercise or nutrition recommendation engine — [1] and [3] use static or periodic biomarker inputs rather than continuous multi-modal fusion.

6.5 Gap 5: AI Trustworthiness in Nutrition Advice Remains Open

[3] explicitly raises the concern that AI-generated nutrition advice lacks expert-guideline alignment verification at scale, and [4] identifies genetic variation and long-term effects as unresolved. For a system like FitTrack Pro that will generate diet plans for users who may have undisclosed health conditions, this is a direct and unresolved safety-relevant gap, not just an accuracy concern.

7. Future Research Directions

Direction	Rationale
Joint exercise-diet optimization	Extend [1]'s contextual-bandit exercise engine and [3]'s generative diet engine into a single multi-objective model that recommends workout and meal plans jointly against one user profile, closing Gap 1.
Mandatory subgroup fairness reporting	Adopt [7]'s fairness-gap methodology as a standard evaluation step for any personalization system, including exercise and wearable-monitoring models, closing Gap 2.
Longitudinal FitTrack Pro pilot	Design a 6–12 month pilot (beyond [1]'s 4 weeks and [4]'s few-month norm) tracking whether personalized recommendations sustain behaviour change, directly addressing Gap 3.
Continuous-fusion-to-recommendation loop	Feed [8]'s real-time multi-modal vital-sign fusion directly into the recommendation engine so workout intensity adapts in real time to detected stress or cardiac signals, closing Gap 4.
Guideline-verified generative nutrition advice	Build on [3]'s loss-function alignment approach and add an explicit clinical-guideline verification layer before any diet plan is shown to a user with a flagged health condition, addressing Gap 5.
Genetic and long-term data integration	Following [4]'s finding that only 3 of 67 studies used genetic data, future FitTrack Pro iterations could offer optional genetic-data integration for users who opt in, while defaulting to the safer non-genetic model for others.

8. Comparative Literature Survey Table

The table below consolidates all eight reviewed papers into the format required for this assessment: author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Mehrabadi et al., 2024	Personalized exercise recommendation	RL / contextual multi-armed bandit	Smartphone+smartwatch apps, n=20	P<.001 increase in exercise duration; 4.31/5 satisfaction	Small, single-demographic pilot (20 female students)	No cross-demographic validation of the RL approach
Fang et al., 2024	Personalized exercise goal setting	Knowledge-based + fitness/fatigue evaluation model	Exercise performance & fitness profile data	Adaptive goals outperform fixed daily targets	No large-scale quantitative accuracy reported	Lacks empirical validation at population scale
Papastratis et al., 2024	AI-based diet/meal recommendation	Variational autoencoder + ChatGPT	3,000 virtual + 1,000 real profiles	High accuracy, variety, and generalization	Guideline-alignment not independently expert-verified	Trustworthiness of AI nutrition advice unresolved
Tsolakidis et al., 2024	State of AI/ML in personalized nutrition	PRISMA systematic review, 3 databases	67 studies (from 3,052 screened)	Recommendation systems dominate the field	Only 3/67 studies use genetic data	Limited genetic/precision personalization
Theodore Armand et al., 2024	AI/ML/DL applications in nutrition broadly	Hybrid SLR + PRISMA review	Field-wide synthesis, Nutrients journal	Broad versatility across nutrition sub-domains	Review is descriptive, not a new model	No unifying framework proposed across sub-domains
Shajari et al., 2023	AI-based wearable sensor technology	Structured hardware + AI review	Field-wide synthesis, Sensors journal	Strong potential for personalized monitoring	Accuracy/diagnostic-precision limits remain	Biosensor reliability not yet production-grade
Chen & Wang, 2025	Population-scale personalized fitness ML	XGBoost / Decision Tree / ANN + k-means	NHANES 2017–2020 + BRFSS, ~15,000 participants	MeanIoU 0.789, fairness gap <0.05, best-in-class MAE	Cross-sectional data; slightly lower accuracy for minorities	No longitudinal validation of behaviour change
Ma'aitah et al., 2026	Real-time multi-modal wearable health monitoring	CNN+LSTM fusion (MAHN), edge-cloud hybrid	3,247 participants, 8-month multi-sensor data	94.3–98.7% accuracy across 5 monitoring tasks	Continuous fusion implies ongoing compute/connectivity cost	Not yet linked to a downstream recommendation engine

9. Conclusion

This literature survey reviewed eight peer-reviewed papers published between 2023 and 2026, covering the three technical pillars of the FitTrack Pro problem statement: personalized exercise recommendation, AI-driven nutrition recommendation, and wearable-based health monitoring. Across these papers, machine learning and deep learning approaches — reinforcement learning, generative models paired with large language models, gradient-boosted ensembles, and CNN-LSTM sensor fusion — consistently outperform static, one-size-fits-all guidance, with reported gains ranging from statistically significant exercise-duration increases to detection accuracies above 94% for physiological monitoring tasks.

At the same time, the comparative analysis exposed real, evidence-backed limitations: small or narrow-demographic pilot studies, short evaluation windows, incomplete fairness auditing, and — most directly relevant to FitTrack Pro — no single reviewed system that jointly recommends both exercise and diet from one unified user profile, or that closes the loop from real-time wearable sensing directly into a recommendation engine. These five research gaps, and the corresponding future directions proposed in Section 7, give FitTrack Pro's continued development a literature-grounded roadmap rather than a purely intuition-driven one.

References

References are listed in IEEE numbered style, corresponding to the citation numbers [1]–[8] used throughout this report.

- [1] M. Asgari Mehrabadi, E. Khatibi, T. Jimah, S. Labbaf, H. Borg, L. Narvaez, P. Pimentel, A. Turner, N. Dutt, Y. Guo, and A. M. Rahmani, “PERFECT: Personalized Exercise Recommendation Framework and architECTure,” *ACM Transactions on Computing for Healthcare*, vol. 5, no. 4, Art. 34, Nov. 2024, doi: 10.1145/3696425.
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- [5] T. P. Theodore Armand, K. A. Nfor, J.-I. Kim, and H.-C. Kim, “Applications of Artificial Intelligence, Machine Learning, and Deep Learning in Nutrition: A Systematic Review,” *Nutrients*, vol. 16, no. 7, Art. 1073, Apr. 2024, doi: 10.3390/nu16071073.
- [6] S. Shajari, K. Kuruvashetti, A. Komeili, and U. Sundararaj, “The Emergence of AI-Based Wearable Sensors for Digital Health Technology: A Review,” *Sensors*, vol. 23, no. 23, Art. 9498, Nov. 2023, doi: 10.3390/s23239498.
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